# JIN SONG

Email: JinSongResearch@outlook.com; Tel: +86 182 0196 3248

## EDUCATION

**Nanjing University of Posts and Telecommunications** *Nanjing, China* Sept/2022 - Jun/2026
**Major**: Information Security (Bachelor of Science in Engineering); **GPA**: 3.92/5.0; **WES GPA**: 3.73/4.0
***Youth Pioneer Service Star*** (Dec 2024)
***School Club Contribution Award*** (Sept 2024)
***Individual Course Award*** (Sept 2023)

## PUBLICATIONS

Yuan Yao, **Jin Song**, Jian Jin. ***Hashed Watermark as a Filter: A Unified Defense Against Forging and Overwriting Attacks in Neural Network Watermarking*** (AAAI 2026 Poster)
Yuan Yao, **Jin Song**, Huixia Li, Tongtong Yuan, Jiaqi Wu, Yu Zhang. ***Semi-Supervised Noise Adaptation: Transferring Knowledge from Noise Domain*** (ICLR 2026 under review)
Yuan Yao, Lixu Wang, Jiaqi Wu, **Jin Song**, Simin Chen, Zehua Wang, Zijian Tian, Wei Chen, Huixia Li, Xiaoxiao Li. ***FedRE: A Representation Entanglement Framework for Model-Heterogeneous Federated Learning*** (CVPR 2026 under review)
Yi Yang, Xueqi Li, Yiyang Chen, **Jin Song**, yihan wang, Zipeng Xiao, Jiadi Su, You Qiaoben, Pengfei Liu, Zhijie Deng. ***Mantis: A Versatile Vision-Language-Action Model with Disentangled Visual Foresight*** (CVPR 2026 under review)

## RESEARCH EXPERIENCES

**Mantis** Jun/2025-Present

***Supervisor***: *Zhijie Deng, Yi Yang*

- Developed a vision-language-action (VLA) model that leverages future visual dynamics to guide action learning.
- Proposed Disentangled Visual Foresight (DVF), a diffusion module that learns compact motion representations from inter-frame visual dynamics and uses them to guide action generation.
- Designed a progressive multimodal training strategy that incrementally integrates vision, action, and language supervision, effectively preserving language understanding and reasoning capabilities during robot policy learning.
- Pretrained the model on 220K human manipulation videos, 76K robot demonstrations, and 38 multimodal datasets, enabling the model to interpret diverse instructions and execute corresponding behaviors.
- Achieved 96.7% success on the LIBERO benchmark with faster convergence than prior VLA methods based on visual foresight, demonstrating the efficiency of the DVF design.
- Validated strong instruction-following ability and out-of-domain generalization with real-world robot experiments, including tasks requiring world knowledge and complex reasoning.
- Proposed an Adaptive Temporal Ensemble (ATE) inference strategy that dynamically adjusts ensemble strength based on required motion stability, reducing inference calls by up to 50% while maintaining task success rates.

**Federated Representation Entanglement (FedRE)** Jan-May/2025

***Supervisor***: *Yuan Yao*

- Proposed FedRE, a federated learning framework that aggregates client knowledge at representation level.
- Formulated entangled representations that combine all local representations from a client into a single cross-category vector using normalized random weights, enabling client-level knowledge sharing.
- Coupled entangled representations with parallel label encodings to provide soft supervision for global training..
- Introduced random weight resampling across communication rounds to diversify supervision signals, reducing optimization sensitivity to specific entanglement configuration and improving generalization.
- Reduced communication overhead by uploading only one entangled representation from each client at every communication step, achieving over 10× lower upload cost than full representation on TinyImageNet.
- Enhanced privacy against representation inversion attacks by entangling representations from multiple categories, reducing reconstruction quality to a PSNR of 9.66, compared to 12.89 for raw representations.
- Demonstrated consistent and stable performance gains over prototype-based baselines across multiple benchmarks under both practical and pathological heterogeneous settings. (up to ~12% absolute accuracy improvement)
- Validated robustness across diverse architectures, including CNNs and Transformers, and partial client participation settings, demonstrating scalability to realistic federated applications.

**Noise Adaptation Framework (NAF)** Sept/2024-May/2025

***Supervisor***: *Yu Zhang, Yuan Yao*

- Formulated the Semi-Supervised Noise Adaptation (SSNA) problem, aiming to introduce synthetic noise as a surrogate source domain to improve the generalization of target domain under extremely limited supervision.
- Derived a generalization bound that theoretically characterizes how noise can reduce target error by controlling cross-domain distribution divergence in a shared representation space.
- Identified that label alignment induces transferable discriminative structure in the noise domain, enabling effective knowledge transfer despite the absence of semantic meanings.

- Proposed the Noise Adaptation Framework (NAF), which explicitly tightens the target generalization bound by jointly optimizing target loss, noise loss, and domain alignment objectives.
- Designed Negative Domain Similarity (NDS), an alignment mechanism that measures discrepancy between the target and noise domains via computing the cosine similarity of their global means.
- Achieved substantial improvements over Empirical Risk Minimization (ERM) across multiple benchmarks, and additionally demonstrated feasibility on large-scale datasets. (up to ~15% absolute accuracy improvement)
- Demonstrated that NAF improves multiple semi-supervised learning methods as a plug-in module, confirming its generality beyond specific architectures or losses. (up to ~20% absolute accuracy improvement)

**NeuralMark** Dec/2023-Sept/2024

***Supervisor**: Yuan Yao*

- Proposed NeuralMark, a model watermarking method that embeds ownership information into model parameters.
- Designed a hashed watermark filter that maps the secret key to an irreversible binary watermark via SHAKE-256 and uses it as a parameter selection criterion, effectively preventing gradient-based reverse-engineering attacks.
- Introduced a multi-round filtering strategy that reduces parameter overlap between different watermarks, providing embedding isolation and substantially increasing resistance to adaptive overwriting attacks.
- Incorporated average pooling over filtered parameters to enhance watermark robustness against parameter perturbations, such as fine-tuning and pruning, without degrading task performance.
- Formulated a joint training objective that embeds NeuralMark through a lightweight embedding loss alongside the main task loss, enabling seamless integration without modifying model architectures.
- Established a theoretical security boundary for watermark verification, quantifying the probability of successful forgery and deriving a statistically grounded detection threshold for ownership confirmation.
- Conducted extensive experiments across 13 architectures, spanning 5 image benchmarks and 1 text benchmark, showing negligible degradation in task performance while maintaining reliable watermark detectability.
- Empirically validated strong robustness under multiple adversarial conditions.

## PROFESSIONAL COMPETITIONS

**The Global Multimedia Deepfake Detection** Jun/2024

*Individual; **Rank 90 / 706; Top 12.8%***

- Performed offline augmentation for the minority positive samples and online augmentation for the majority negative samples, thereby improving the diversity of the official unbalanced dataset.
- Tried various image classification architectures and hyper-parameters for 1 epoch and chose EfficientNet-B1 as the final model due to memory constraints of Kaggle.

**The 2nd World AI4S Prize-Logical Reasoning Track Evaluation of Complex Reasoning Ability** Jun/2024

*Individual; **Rank 129 / 772; Top 16.7%***

- Utilized Test Time Scaling (Majority Vote) and Prompt Engineering to improve the accuracy of Qwen2-7B-Instruct.

**Alibaba Cloud Tianchi College Student Competition** Sept/2024

*Individual; **Rank 74 / 449; Top 16.5%***

## PATENTS

A federated learning method, system, device, medium, and product with model heterogeneity, **CN202510920651.0**

A weight-based neural network watermarking method, system, device, and medium, **CN202510758878.X**

## EXTRACURRICULAR ACTIVITIES

**Academic Tours & Conferences** Sept/2022-Present

- Distinguished Group Leader of the Summer Study Tour at NTU (2024)
- AWS Summit Shanghai (2024)
- TA at Datawhale AI Summer Camp (2024)
- Trusted Federated Learning Winter Camp - Westlake University (2023)

**Student Science Association** *President 60hrs /w* Sept/2023-Present

- Examiner of Institute Theoretical Application Logic Competition (2024)
- Judge of the Student Science Association Freshmen Recruitment (2023)

**Tennis, Video Game, Music, Photography, Traveling** Present

## PROFESSIONAL SKILLS

**Programming Languages**: C, C++, Python (3yrs)
**Tools**: Latex (1yrs), Markdown (3yrs)
**Frameworks**: PyTorch (2yrs)
**Operating Systems**: Linux (1.5yrs), Windows
**Languages:** Native speaker of Chinese, Fluent in English